

MediRoute: Autonomous Medical Supply Delivery Robot for Hospitals

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Project Abstract

MediRoute is an autonomous medical supply delivery robot designed to safely and efficiently transport medicines between hospital departments and wards. The system uses line-following navigation, obstacle detection, a secure medicine storage unit, and destination selection to automate internal medical deliveries. A web-based monitoring dashboard enables staff to place orders, track delivery status, confirm receipt, and manage emergency medical requests in real time.

Problem Statement

Hospitals often face delays and inefficiencies in manually transporting medicines and medical supplies between departments and wards, increasing workload and the risk of delivery errors.

Need for the Project

There is a need for an automated and reliable system to deliver medicines and medical supplies within hospitals, reducing manual workload, delivery delays, and the possibility of human errors.

Project Objectives

- To develop an autonomous robot for transporting medicines and medical supplies within hospitals.
- To implement line-following navigation and obstacle detection for safe and reliable movement.
- To provide a web-based monitoring system for placing orders, tracking deliveries, and confirming receipt.
- To reduce manual workload, delivery time, and human errors in internal hospital medicine transportation.

Expected Learning Outcomes

- Gain practical knowledge in robotics, automation, and autonomous navigation.
- Develop skills in sensor integration, line following, and obstacle detection.
- Learn to design and implement a web-based monitoring and control system.
- Develop experience in mechanical design, system integration, testing, and troubleshooting.

System Requirements

Hardware

- Arduino Uno, 3-channel IR sensor array, L293D motor driver, 2 DC geared motors, breadboard, battery pack, chassis

Software

- Arduino IDE, Embedded C/C++, Python, Flask, HTML, CSS, JavaScript, Supabase, and Netlify.

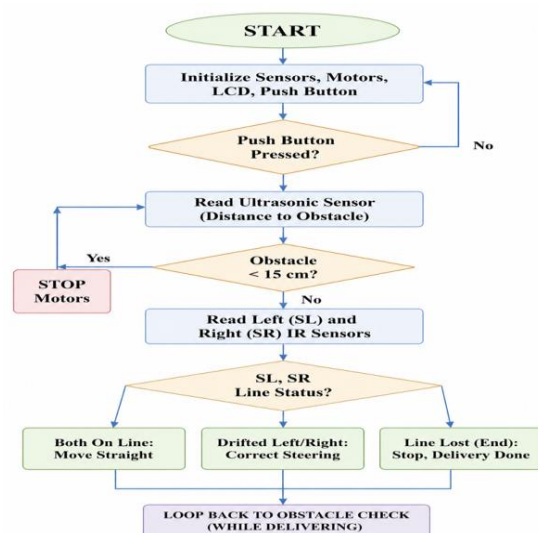
Methodology / Workflow

Delivery Request on Web Dashboard → Order Processing → Medicine Loading into Robot → Destination Selection → Robot Follows the Line to the Destination → Medicine Delivery → Click “Return” → Robot Returns to the Starting Point

System Architecture

Web Dashboard → Order Processing → Medicine Loading → Destination Selection → Arduino Uno Controller → IR Sensor-Based Line Following → Motor Driver & DC Motors → Destination Arrival → Medicine Delivery → Return Command → Robot Returns to Starting Point → Web Dashboard Status Update

Block Diagram



Project Development Process

Requirement Analysis → System Design → Component Selection → Circuit & Robot Development → Software/Web Dashboard Development → Hardware–Software Integration → Prototype Assembly → Testing & Debugging → Final Validation

Implementation

Developed the robot using Arduino Uno, IR sensors, L293D motor driver, and DC geared motors; integrated the medicine storage unit and web dashboard to process delivery requests, select destinations, control line-following movement, deliver medicines, and return the robot to the starting point.

Algorithms / Models Used

- IR-Based Line Following Algorithm
- Motor Speed Correction Algorithm

Dataset / Input Data (if applicable)

No fixed dataset — the system uses real-time IR sensor readings, calibrated using ambient-light samples before each run.

Experimental Setup

Tested on a 3 m track with black tape on white, including straight segments, curves, and a 90° intersection under normal indoor lighting.

Results

The robot tracked the line reliably across all segments, with under 1 cm deviation on curves and consistent recovery at intersections.

Performance Evaluation

Evaluated on tracking accuracy, sensor response time, and average speed — all showed consistent real-time correction.

Testing and Validation

Validated across multiple track layouts, repeated runs, and varying lighting to confirm repeatable line-following.

Challenges Encountered

- Sensor readings were sensitive to ambient light, causing false detections.
- Slight motor speed mismatch caused drift on straight runs.

Solutions Implemented

- Added a calibration routine to adjust sensor thresholds per run.
- Applied software speed compensation to balance the motors.

Innovation / Novel Features

A lightweight threshold-calibration routine lets the robot adapt to lighting changes without hardware modification.

Applications

- Automated material handling in warehouses and factories
- Educational platform for embedded systems and robotics
- Hospital Medicine Delivery

Future Scope

Extend with obstacle-avoidance sensors, wireless control, and ML-based path prediction for unmarked environments.

Project Outcomes

Delivered a working, low-cost line-following robot demonstrating reliable autonomous navigation using Arduino-based sensor and motor control.

Skills Acquired

- Embedded programming and circuit design
- Teamwork and problem-solving under constraints

References

- Design and Construction of Line Following Robot using Arduino, *International Journal of Trend in Scientific Research and Development (IJTSRD)*, Vol. 3, Issue 4, May-Jun 2019.
- Path Following Robot using Arduino and IR Sensors, *International Research Journal of Engineering and Technology (IRJET)*, Vol. 11, Issue 5, May 2024.

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DEMO VIDEO



PROJECT REPORT

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